	AzureSky		
	AS003IW	PROCEDURE ON THE DETERMINATION OF ASH CONTENT IN HARD COAL	
	Compiled by:	Sifiso Nyambi- Laboratory Manager	
	Approved by:	Nonhlanhla Msibi- Operations Manager	

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1. Introduction

When coal is burnt in the presence of oxygen it leads to the formation of a residue called ash. Ash is an incombustible, and undesirable matter. This is because ash causes slagging and fouling in coal-fired boilers. Electricity is generated by Power Stations through coal usage. The energy released by coal during combustion is used to heat water, creating steam, this steam under extensive pressure is then used to turn turbines. The turbines are connected to generators which produce electricity **(see Faraday's law of electromagnetic induction for a deeper understanding)**.

Note: Coal with high ash content have low energy output (i.e. they have low calorific value).

There are two types of ash analysis that can be used within the laboratory, ISO ash and quick ash:
ISO Ash is characterised by the steady increase in temperature of the furnace (termed temperature ramping), on the other hand, quick ash is characterised by the rapid incineration (temperature remains at 815°C throughout the duration of analysis). Although one might get similar results from both methods, it is extremely important to note that coal samples behave differently under different temperatures. For example, a sample will have a higher sulfur content in its ash if it were analysed by quick ash than if it were analysed by ISO ash. The gradual increase in temperature for ISO ash allows for certain compounds to decompose at lower-to-mid temperatures and be released from the coal as sulfur oxides, that does not happen in quick ash. More sulfur content means the coal is heavier, thus there could potentially be significant errors in the percentage determination of ash results.

NOTE: ONLY PERSONNEL WHO ARE AUTHORIZED AND DECLARED COMPETENT ARE TO PERFROM LABORATORY ACTIVITIES. PERSONNEL NOT DECLARED ARE TO STRICTLY WORK UNDER SUPERVISION OF A LAB MANAGER/SUPERVISOR OR COMPETENT PERSONNEL.

2. Scope

To determine the ash content of coal samples.

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3. Principle

Test portion of coal sample is heated in air from ambient temperature up to 815±5 ºC and maintained at this temperature until a constant mass is achieved. The percentage of ash content can thus be obtained after complete incineration of sample.

4. Apparatus

4.1. Balance capable of weighing the nearest 0.1mg
4.2. Furnace capable of being heated to 815±5 ºC and maintained at this level until a constant mass in obtained. The ventilation through the furnace shall be such as to give 5-10 air changes per minute, these can be assessed by measurement of airflow in the furnace flue using a pitot static and suitable manometer.
4.3. Crucible derived from porcelain with well that is 8-15 mm deep of such size that the loading can accommodate a coal sample portion not exceeding 150g/cm$^{2.}$
4.4. Desiccator or another enclosed container without a desiccant
4.5. A pair of thongs with a hand shield.

Verification of furnaces are to be done every Monday of every week. Insert the probe into the furnace; Ash furnace readings is taken done as follows: 1st reading at 30 minutes, 2nd reading at 45 minutes and last reading at 1 hour. For volatile furnace, 1st read at 4 minutes after insertion of the external probe and the second and third read at 7-minute intervals consecutively. Apply the correction factors as specified in the calibration certificates of the external thermometers. The principle is to mathematically add the correction factor. This is to be documented in the AS024TF form.

Note: Ensure all equipment and materials used are verified against factory specifications and logged in the equipment register prior to use. Calibrations (where necessary) and verification of equipment are to be attended prior to use. Calibration services shall only be sourced from ISO/IEC 17025 accredited labs. Verification certificates from manufacturers are accepted for new equipment provided to address traceability.(ISO 17025; Clause 6.4.)

5. Preparation of sample

The sample shall be prepared as per the **AS001WI** procedure and conditioned as per **AS002WI**. Ensure that the sample is -212 µm and mix well to homogenise before conditioning it.

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6. Procedure

A- ISO ASH

6.1. Place the empty, clean and dry crucible (4.3) on top of the analytical balance, close the door, allow for the balance to stabilise and press the spacebar key on the keyboard to auto transfer the mass.
6.2. Press the tare button on the balance
6.3. Weigh 1.0000±0.05 g sample into crucible, close the door and allow to stabilise. Press the spacebar on keyboard.
NB: Do not forget to spread the sample evenly across the area of the crucible
6.4. Repeat the above steps for the duplicate sample.
6.5. Carry the tray with sample to the furnace room and place it in front of the ISO furnace you are going to use.
6.6. Open the furnace door and use the pair of thongs to place the sample into the furnace chamber and then close the door.
6.7. Turn on the furnace by pressing and holding the reset button for about 5 seconds. The temperature should start to increase and will reach 500°C within 60 minutes, this temperature shall be maintained for about 30 minutes. Next, the temperature will rise all the way to 815±5 ^{o}C in the next 60 minutes and shall maintain this temperature for 60 minutes.
6.8. Finally, the temperature will gradually decrease (~1 hour) until the chamber achieves an ambient temperature.
6.9. Use the tongs to remove the crucibles from the furnace and place them onto the bench to cool for 10 minutes. Additional 10 minute inside desiccator.
6.10. Weigh the sample by pressing the spacebar key.

B- QUICK ASH

6.1.1. Place the empty, clean and dry crucible (4.3) on top of the analytical balance, close the door, allow for the balance to stabilise and press the spacebar key on the keyboard to auto transfer the mass OR manually insert the mass.
6.1.2. Press the tare button on the balance.
6.1.3. Weigh sample 0.5000<mass<0.2495 g into crucible, close the door and allow to stabilise. Press the spacebar on keyboard.

NB: Do not forget to spread the sample evenly across the area of the crucible.

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6.1.4. Repeat the above steps for the duplicate sample.
6.1.5. Carry the tray with sample to the furnace room and place it in front of the quick ash furnace you are going to use.
6.1.6. Open the furnace door and use the pair of thongs to place the sample into the furnace chamber and then close the door.
6.1.7. Allow the sample to burn for a period of 1 hour. Check if the sample has completely ashed after the hour has elapsed, if not put the sample back into the furnace for an additional 30 minutes.
6.1.8. Take out the sample from the furnace and cool for 10-15 minutes.
6.1.9. Weigh the sample by pressing the space key.

7. Result expression and calculations

M1= Mass of empty crucible
M2= mass of sample
M3= Mass of crucible+ ash

%Ash = $\frac{M3-M1}{M2} X100$

All results are reported are in an air-dry basis

Repeatability criteria:

- Ash content < 10% RL of 0.2%
- Ash content >10% RL 2% of mean
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Reproducibility criteria is 0.3 and 3% of the mean, respectively.

8. References

ISO 1171:2024

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9. Table of Revision

Rev number	Date	Reason for revision
01	02/2026	Under apparatus section, added how verification of furnaces shall be done, its frequency included.

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